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A scoping review of interventions addressing social determinants of health and their influence on opioid use disorder outcomes

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Abstract

Background: Over the past decade, opioid overdose deaths have sharply increased. The Social Determinants of Health (SDOH) framework can guide interventions to improve opioid use disorder (OUD) outcomes. No comprehensive review of SDOH interventions and their impacts on OUD outcomes is available. This scoping review addresses that gap.

Methods: We extracted articles from PubMed, Embase, Web of Science, and Cochrane databases. Interventions were categorized according to SDOH domains: healthcare system, social and community context, neighborhood environment, economic stability, and education. Interventions and OUD outcomes (e.g., treatment initiation, opioid use, overdose) were summarized.

Results: Twenty-seven peer-reviewed studies targeted SDOH domains. The healthcare system domain (37 % of studies) was the most frequently addressed, focusing on provider training, access, and quality-of-care improvements with outcomes like increased initiation of medication for OUD, reduced opioid use, and reduced provider stigma. Community and social context (30 %) interventions included social support programs and community coalitions that reduced opioid

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use, overdose, and community-level stigma. Economic stability (16 %) interventions included employment-based reinforcements and financial incentives to promote abstinence. Neighborhood and physical environment (9 %) interventions included Housing-First initiatives that reduced opioid use. In the education domain (2 %), an early education intervention reduced adulthood OUD risk. Over half of all studies (52 %) used randomized designs; the remainder used quasi-experimental approaches. Gaps included a limited range of SDOH interventions, inconsistent definitions and measurements of SDOH, and a lack of rigorous evaluations.

Conclusion: Future research should harmonize SDOH terminology and metrics, rigorously assess SDOH intervention outcomes, and expand the range of SDOH interventions.

Keywords

Social Determinants of Health; Opioid Use Disorder; Substance-Related Disorder; Health Disparity; United States

1. Introduction

In 2023, approximately 6 million people, representing 3 % of the United States (U.S.) population age 12 years or older, reported an opioid use disorder (OUD) in the past year (Substance Abuse and Mental Health Services Administration, 2024). The toll of opioid-related deaths, which more than tripled from 21,000 in 2010 to over 68,000 deaths in 2020, has disproportionately impacted people of color (Mason et al., 2022; Milam et al., 2021) and people from economically disadvantaged backgrounds (Altekruse et al., 2020; van Draanen et al., 2020). The alarming rate of opioid-related deaths and widening health disparities underscore the need to leverage the Social Determinants of Health (SDOH) framework to understand the conditions that shape the development of OUD and intervene with those conditions to improve OUD outcomes.

SDOHs are defined as “conditions in the environments where people are born, live, learn, work, play, worship, and age” that shape health (Kaiser Family Foundation, 2018; Office of Disease Prevention and Health Promotion, 2024). SDOH domains—education access and quality, economic stability, healthcare access and quality, neighborhood and built environment, and the social and community context—are unevenly distributed in society and result in unequal exposure to factors associated with a variety of health conditions, including OUD (Cockerham et al., 2017; KFF, 2018; National Institute on Drug Abuse, 2023a; Yelton et al., 2022). SDOH factors influence the development, persistence, and outcomes of OUD, influence access to quality healthcare and licensed medical professionals, and affect OUD outcomes, including initiation, remission, and mortality (Abraham et al., 2013; Chang et al., 2022; Han et al., 2022; Hansen et al., 2022; Hser, 2007; Jin et al., 2017; Lin et al., 2024; Lopez et al., 2021; McKenzie et al., 2016; Rangachari et al., 2022; Whitesell et al., 2013; Whittle et al., 2019). See Table 1 for examples of SDOH disparities and OUD risk.

Understanding interventions that target SDOH and assessing their association with OUD outcomes can provide critical insights to meet the needs of individuals at risk of developing OUD or who are diagnosed with OUD. However, comprehensive reviews that explore how SDOH interventions affect OUD outcomes are sparse. For example, Sugarman and

colleagues (2020) found that evidenced-based interventions addressing OUD and SDOH were lacking among people experiencing incarceration (Sugarman et al., 2020). Formosa and colleagues (2019) summarized the evidence of SDOH interventions for patients experiencing homelessness yet limited their review to emergency department (ED) settings and did not specifically focus on OUD (Formosa et al., 2019).

To address the gaps in knowledge about SDOH and their influence on OUD-related outcomes, the present work conducted a scoping review (Munn et al., 2018) to examine SDOH interventions in the context of OUD across multiple settings and populations. Specifically, this review uses an SDOH framework to identify, evaluate, and synthesize the current knowledge of SDOH interventions and their impacts on OUD outcomes. To this end, the research questions are: Toward improving OUD outcomes, what interventions have been implemented to change factors across the SDOH domains (i.e., interventions on education, economic stability, healthcare access and quality, neighborhood or physical environment, and social and community context, as well as economic stability)? What are the outcomes of these SDOH interventions? What are the strengths and weaknesses in SDOH-focused interventions? This review is timely as the 2022–2026 NIDA strategic plan calls for research on the impact of SDOH on substance use, treatment utilization, outcomes, and recovery, as well as research on interventions that target SDOH factors to help achieve more equitable OUD outcomes (National Institute on Drug Abuse, 2023a; Volkow, 2023).

2. Methods

We defined an SDOH intervention as a program or policy seeking to improve OUD outcomes by addressing the social, economic, and environmental conditions that contribute to OUD and its consequences. Using the KFF and Healthy People 2030 frameworks (Kaiser Family Foundation, 2018; Office of Disease Prevention and Health Promotion, 2024), we summarized interventions by SDOH domains: (1) education; (2) healthcare system; (3) economic stability; (4) neighborhood and physical environment; and (5) community, safety, and social context. SDOH are often conceptualized as upstream factors (e.g., improving social conditions through addressing root causes at the society or community level). However, to provide a comprehensive review of SDOH interventions, similar to work by others (Whitman et al., 2022), we did not exclude interventions at the mid-level (e.g., individual-level needs from broader societal conditions) or downstream (e.g., address immediate social needs at the individual level).

This literature review was also guided by elements of the Preferred Reporting Items for Systematic Reviews and Meta-analyses 2020 guidelines. The protocol is pending review with Open Science Framework (<https://osf.io/6b7q4>).

2.1. Search strategy

Informational databases included PubMed, Embase, Web of Science, and Cochrane Review. The search included articles published between 2013 and 2024, focusing on studies conducted in the U.S. The cutoff day of the search was September 10, 2024. The timeframe sought to capture the most recent shifts in substance use patterns and contemporary developments in OUD research and intervention effectiveness. The search strategy combined

the search terms from key concepts, including intervention studies (e.g., intervention trial, randomized controlled (RCT) trial, pragmatic clinical trial, evaluations) and opioid use (e.g., opioid or heroin use, opioid-related disorders, etc.) as well as SDOH terms (i.e., social determinants, or specific factors within education, health care and quality, economic stability, neighborhood or physical environment, and social and community context domains of the SDOH). The search terms (see Appendix) were reviewed by a librarian and included MeSH, title, abstract, and keywords to ensure thoroughness. A hand search conducted on Google Scholar supplemented the results. Records were imported into Mendeley Reference Manager for deduplication.

2.2. Screening criteria

After removing duplicates (N = 468), 676 articles potentially met the initial eligibility criteria. Abstracts and titles were screened using Rayyan, a cloud-based web app developed for systematic reviews (Ouzzani et al., 2016). Exclusion criteria included 1) studies conducted outside of the United States; 2) article not in English; 3) full article not available; 4) not human subjects; 5) outcomes unrelated to opioid use or OUD, not disaggregated, or were related to pain management or surgery; 6) no intervention described (e.g., cross-sectional studies describing associations with SDOH); 7) intervention not targeting SDOH domains or factors; 8) wrong publication type (e.g., protocols, reviews, editorials, opinions) and 9) sample size was less than 20 individuals.

2.3. Data extraction and analysis

From abstract and title screening, 62 studies potentially met eligibility criteria (Fig. 1). Full texts were imported into Atlas.ti (Version 22.0) for content analysis. Initial codes were developed to characterize SDOH interventions (guided by the KFF SDOH framework), study design, targeted population, setting, intended outcomes (e.g., opioid use, access to treatment, overdose prevention) and main findings. The priori codes evolved throughout the coding process to capture additional SDOH elements and emerging themes. Five Ph.D. or Master-level researchers with research experience in OUD and SDOH conducted screening and coding. Thirty-five articles were excluded due to not having an SDOH-related intervention, wrong study design, outcome not related to opioid use or OUD, full text unavailable, not in the U.S., or the study population was under 18 years old. We categorized 27 articles according to SDOH domain, study design, SDOH intervention strategy location, setting, population, and outcomes. Two coders reviewed each article at each phase, and coding discrepancies were resolved through consensus meetings.

3. Results

3.1. Study characteristics

Two-thirds (67 %) of the articles were published in 2020 or later; the rest were published earlier. Over half (52 %, n = 14) of the articles included randomization, while 48 % (n = 13) were quasi-experimental designs (Table 2). Most studies (81 %) focused on individuals using opioids or with OUD. Interventions were primarily conducted in healthcare settings (52 %), with fewer in community (44 %), and school settings (4 %). Common study locations included Massachusetts (15 %) and New York, North Carolina, Maryland, Ohio,

Pennsylvania and Washington (each 7 %). More than a third (37 %) of studies focused on the SDOH domain of healthcare quality and systems, followed by 30 % addressing community and social context, 16 % focusing on economic stability (9 % on neighborhood and physical environment, and only one article explored the education domain.

3.2. Healthcare system domain

Over a third of the studies of interventions that focused on the healthcare system domain (38 %) included randomization in their design. Interventions in this domain aimed to enhance healthcare quality and access through changing provider knowledge and practice, implementing clinical guidelines for opioid medications, coordinating access to care, and reducing stigma. However, it is important to note that some of these interventions were both upstream SDOH (social/structural influences) and direct medical care interventions. See Tables 3, 4 and 5.

3.2.1. Provider knowledge and practice: Whiteside and colleagues (2021) piloted an intervention that included case management, bedside linkages to care, overdose medications, and provider-level OUD education. While this intervention focused on access to care through case management and linkages, it also involved a clinical care component (overdose medication, naloxone). The number of days of illicit opioid use decreased at the 1-month and 3-month follow-up time points for both groups (Mean = 12 vs. 9 days). At 3 months, intervention participants reported 3.5 fewer days of heroin use compared to participants in TAU, although this difference was not statistically significant (Whiteside et al., 2021). McCarthy et al., (2019) sent reminders to ED-based providers to counsel patients about safe usage of opioid medications to prevent opioid misuse. This study produced mixed results: the survey data indicated consultation increased the likelihood of safe opioid use compared to patients in TAU (aOR 2.46, CI 1.19, 5.06) but not when examining returned opioid use diaries.

Two studies included interventions to promote adherence to opioid prescribing guidelines. Quanbeck et al. (2018) conducted a randomized controlled trial with a matched-pair design to implement a systems consultation strategy in eight community-based primary care clinics. This strategy included audit and feedback, academic detailing (face-to-face education), and external facilitation (process management support) to enhance adherence to opioid prescribing guidelines. The authors found intervention group reductions from pre- to post-test in the proportion of patients with three or more opioid prescriptions in the past 3months (but not when compared to TAU (slope: 0.0001, CI: - 0.0001, - 0.0002) but no significant differences compared to TAU (Quanbeck et al., 2018). Another RCT study successfully improved the quality of care by reducing over-prescribing of prescription opioids to prevent misuse (Deshazer et al., 2020).

3.2.2. Stigma in healthcare settings: Howard and Clark (2017) developed a provider training intervention using the Interpersonal Shared Decision-Making model aimed to reduce provider bias against pregnant women who use substances through a training intervention that covered gaps in treatment options, child welfare laws, universal verbal screening, and community resources. They reported a 73 % decrease in the proportion

of providers agreeing that “women with substance use problems should not parent” (Howard and Clark, 2017). Hershberger and colleagues’ (2022) randomized study tested an online virtual reality simulation of a healthcare encounter to reduce stigma among health professionals. The simulation included immersive scenarios featuring a pregnant woman with OUD and evidence-based habit-breaking strategies such as perspective taking and individuation to reduce prejudice. Although the mean difference in overall bias scores pre-test/post-test was not significant for pregnant women using opioids, providers showed significant improvements in reduced frustration and increased compassion (mean differences of $-.45$ and $-.31$, $p < .05$) (Hershberger et al., 2022).

3.2.3. Quality of care and access: Molfenter et al. (2019) implemented a multi-organizational learning collaborative to reduce reimbursement and access barriers to MOUD as buprenorphine (BUP). This intervention incorporated Network for the Improvement of Addiction Treatment (NIATx) learning sessions and coaching for payers and treatment agencies to make organizational changes facilitated by implementation teams comprised of representatives from Ohio’s Alcohol, Drug Addiction, and Mental Health Service board and treatment agency staff. The teams addressed workforce, operational, and access-related barriers. For example, to address operational barriers to identify and support patients eligible for BUP, NIATx learning sessions and coaching were used to develop standardized screening workflows, implement diversion prevention protocols, and connect patients to support groups accepting MOUD. As another example, the intervention supported prescriber capacity by recruiting waived clinicians, encouraging waiver applications among existing staff, and maximizing the use of available prescriber slots. The intervention led to a three-fold increase in BUP use among OUD patients (from 9.5 % to 28.5 %), compared to a smaller increase in the comparison group (from 10.9 % to 19.6 %), with an adjusted difference-in-differences of 10 % (95 % CI [9.9 %, 10.7 %], $p < .001$). Hagedorn and colleagues (2022) used block randomization to implement quality improvement interventions in eight Veterans Health Administration facilities. The intervention tested an implementation team to increase MOUD provision. The intervention assessed local barriers and facilitators, formed local implementation teams and action plans for training/education, and provided ongoing coaching and consultation. The proportion of patients with OUD receiving MOUD increased from an average of 18 % at baseline to 30 % one year later.

Ly and colleagues (2024) evaluated the impact of an innovative pharmacist-led pilot service in providing access to BUP treatment to individuals residing in Permanent Supportive Housing and receiving support for their wellness, quality of life, and housing retention. The intervention improved treatment adherence among intervention participants; 37 % post- vs. 3 % pre-intervention participants achieved 80 % of prescription days filled. (Ly et al., 2024).

Gryczynski et al. (2021) conducted an RCT of patient navigators (NavSTAR) provided case management, advocacy, service linkage, and motivational support to address medical and basic needs (housing, food, clothing, etc.). NavSTAR patients had higher rates of medication for OUD (MOUD) treatment entry within 3, 6 and 12 months (68 % vs 40 %; 83 % vs 49 %; 87 vs 67 %) and lower rates of positive opioid drug test within 6 months (68 % vs 74 %) compared to TAU. Conversely, to improve quality care, Suzuki et al. (2023) conducted an RCT of a peer recovery coach intervention with hospitalized patients

with OUD, involving provision of a relapse prevention plan, contact for six months post-discharge, encouragement of medication for MOUD adherence, and identifying linkages to recovery resources. However, there were no significant differences between the TAU group's and the intervention group's retention on MOUD (38.5 % vs 41.7 %), hospital readmissions (46.2 % vs 41.2 %), or time to treatment discontinuation (logrank $p = 0.92$) and readmission (logrank $p = 0.85$) (Suzuki et al., 2023).

3.3. Community and social context

Thirteen (30 %) of the 27 SDOH studies targeted community and social context. Interventions included social integration via peer social supports, utilization of community coalitions to coordinate community-wide changes, or parenting skills to improve the family social context. Three studies (23 %) within this domain included randomization. See Tables 3 and 4.

3.3.1. Peer social support and recovery: Hyde and colleagues (2022) piloted a post-incarceration peer-based intervention to promote engagement in social support, substance use, and mental health services among veterans. Peers offered socioemotional support, referrals, role modeling of life skills, and they attended participants' medical appointments. The intervention was associated with higher linkage and engagement in substance use services (86 % vs. 19 %) and other specialized care for veterans returning to the community after incarceration, suggesting better social integration than the comparison group (Hyde et al., 2022). Crowthers and colleagues (2022) examined a peer recovery coaching intervention that addressed relationships and social support and social determinants like economic stability (e.g., financial independence and employment), education, and independence from legal challenges and institutions. Participants showed higher abstinence rates ($X^2=20.0$), but modest non-significant ($p > .05$) increases in social connectedness (i.e., from 90 % to 98 %; $X^2=2.67$).

3.3.2. Community context and coalitions: Community coalitions can be a coordinated approach to reduce risk factors at the community-level and influence local attitudes (Alexandridis et al., 2019). Alexandridis and colleagues (2019) implemented community-coalition interventions across 74 counties in North Carolina. Coalition efforts included community and school-based education and mass media campaigns to increase community awareness about prescription drug overdose prevention, as well as diversion control efforts (e.g., disposing of unused medications and educating law enforcement about opioid analgesics), and changing local policies to distribute naloxone to OUD-affected families, friends, and first responders. In addition to community context, intervention components overlapped with the SDOH healthcare domain (community-wide provider education, hospital ED policy changes, and MOUD treatment expansion). Interestingly, the diversion control component was linked to increased opioid analgesic dispensing (IRR: 1.06), and other strategies did not reduce prescribing ($p > .05$). Community-wide treatment expansion had no association with changes in BUP utilization (IRR: 0.98) (Alexandridis et al., 2019). Additionally, a grassroots coalition held forums that increased awareness of opioid-involved overdose and resulted in community-level goals to address the OUD crisis (e.g., creation of a local coalition to focus on addressing and prevention of substance

use) (Palombi et al., 2019). The Communities That HEAL (CTH) intervention, a multi-site, community-level, cluster-randomized waitlist-controlled trial, seeks to reduce opioid overdose deaths through evidence-based practices, community engagement, and health communication campaigns in 67 highly impacted communities (Davis et al., 2024; Glasgow et al., 2024). Wave 1 results show a 25 % increase in access to therapies for OUD compared to wait-list controls (Glasgow et al., 2024) and a larger decrease in perceived community stigma toward people treated for OUD (AMC -3.20) and MOUD (AMC -0.33) than in wait-list controlled communities (AMC -0.18 and AMC 0.11) (Davis et al., 2024).

3.3.3. Family context: Family interactions fall within the SDOH domain of community and social context and can significantly impact an individual's health. Godwin and colleagues found lower probability of opioid use later in life among participants who received a comprehensive childhood intervention that targeted, among other needs, parents' parenting skills and behaviors (Godwin and Conduct Problems Prevention Research Group, 2020). Schaeffer and colleagues (2021) conducted an RCT of a family intervention to improve parenting skills, reduce physical interpersonal violence, substance misuse, and child maltreatment, and to improve the overall family context. Parents showed a significant decrease in days of opioid use over time (β : -0.34, $p < .05$) and were more likely to report total past-month abstinence from all drugs (β : -0.06), but the difference was only marginally significant for past-month opioid abstinence (β : -1.45; $p = 0.055$) (Schaeffer et al., 2021).

3.4. Economic stability

Interventions within the economic stability domain ($n = 7$; 16 %) included employment-based reinforcements, employment services via peer navigators and coaches, or financial incentives to bolster economic stability. Over half of the studies in this domain used a randomized design (57 %). See Tables 3, 4 and 5.

3.4.1. Employment services and wage supports: Holtyn and colleagues (2014) assessed whether a therapeutic workplace that offers an employment-based incentive program (contingency-management-based therapeutic workplace) could foster enrollment in methadone treatment and drug abstinence among out-of-treatment patients who inject drugs. Most participants were engaged in methadone treatment during the induction phase (92 %). The therapeutic workplace was a research-based model whereby participants were hired and paid contingent on specific behaviors. For example, training programs helped build skills and were structured to simulate employment while reinforcing treatment engagement and abstinence. In the most intensive intervention condition (compared to work only, or work plus methadone treatment), participants were required to enroll in methadone treatment, attend the workplace, and provide drug-free urine samples to maximize earnings. This intervention resulted in higher percentage of opioid-free urine samples compared to the work-only intervention (75 % vs. 54 %) (Holtyn et al., 2014).

Novak and colleagues (2022) investigated long-term abstinence-contingent wage supplements to promote drug abstinence and employment among unemployed adults enrolled in an opioid agonist treatment program (compared to Treatment as Usual, TAU). All participants were offered an employment specialist throughout the 12-month intervention

period. Participants in the intervention group received stipends for engaging in job-seeking activities and, once employed, wage supplements for hours worked in community jobs, contingent on providing drug-free urine samples. Participants in the intervention group provided a higher rate of opioid- and cocaine-free urine samples than TAU participants (44.1 % vs. 3.6 %) (Novak et al., 2022). As noted earlier, Crowthers and colleagues (2022) included “education services” in their peer recovery coaching study. The authors noted that this intervention included “education and employment services.” Because outcomes were “exclusively related to employment” (i.e., no participants reported education activities at baseline or follow-up), however, we have classified this intervention under the employment domain. Peer recovery was associated with increased rates of being employed ($X^2=19.17$, $p < 0.001$) and, as mentioned above, a higher rate of abstinence ($X^2=20.0$) at the 6-month follow-up compared to baseline.

3.4.2. Financial assistance: Yoo and colleagues (2022) conducted an RCT of low-income mothers with infants to examine if unconditional cash gifts of significant monthly value (\$333) versus a nominal value (\$20) would influence maternal use of alcohol, cigarettes, or opioids. No differences were found in maternal substance use. The effect size estimate for change in opioid use was negative (-0.067) and not statistically significant, suggesting that providing cash gifts to low-income mothers with infants did not increase their opioid use (Yoo et al., 2022).

3.5. Neighborhood and physical environment

Fewer interventions targeted the neighborhood or physical environment ($n = 4$; 9 %), and they were all limited to housing interventions. Interventions included efforts to increase access to affordable housing for those at risk of developing OUD or in recovery from OUD. These interventions were often embedded within more holistic interventions. None included randomization. See Tables 3, 4 and 5.

3.5.1. Housing: Stable and affordable housing can lower the risk of substance use (Substance Abuse and Mental Health Services Administration, 2023a, 2023b). Several articles included Housing First interventions, which provide immediate, permanent housing to individuals experiencing homelessness, coupled with supportive services to address their health and social needs without preconditions such as substance abstinence. Overlapping with the economic stability domain, Kelleher and colleagues (2021) used a Housing-First intervention providing six months of rent, utilities, and OUD prevention services. The intervention covered a housing deposit, application fees (including rental history and credit report), and automatic monthly payments for rent and utilities for 6 months. Despite the high risk of OUD among young adults (over 18 years old) experiencing homelessness, no participants reported opioid use by the 6-month follow-up (Kelleher et al., 2021). Tuten and colleagues (2018) found that participants who accessed recovery housing (i.e., communal, supportive, and supervised substance-free living environments for people recovering from SUDs) during treatment were more likely to be opioid abstinent at 1, 3 and 6 months (β : -1.07 , 1.46 , 1.40 , respectively). Housing First strategies were identified as feasible and effective in reducing opioid use (Davidson et al., 2014; Kelleher et al., 2021). Davidson and colleagues (2014) examined the active participation of people experiencing homelessness,

whose substance use hindered independent housing, in the design and implementation of Housing First. Providers used a Housing First approach for providing client-centered supportive services to improve retention. Clients in programs that included Housing First participants in decision-making were less likely than others to report using stimulants or opioids at follow-up (OR: 0.17). Crowthers and colleagues (2022) used peer recovery coaching, which included the housing services and other supports discussed earlier. The authors found increased rates of housing stability ($X^2=7.14$, $p = 0.008$) and, as previously mentioned, a higher rate of abstinence ($X^2=20.0$) at the 6-month follow-up (compared to baseline).

3.6. Education

Only one of 27 SDOH studies intervened in early education, and it used an RCT to assess intervention effects on opioid use in adulthood (Godwin et al., 2020). See Tables 3, 4 and 5.

3.6.1. Early education: In a randomized pre-test/post-test design to reduce aggression and delinquency in at-risk kindergarteners, Godwin and colleagues (2020) intervened on social and emotional development in early education. They used a comprehensive childhood intervention that targeted academic skills, children's intrapersonal and interpersonal skills, and parenting skills and behaviors through two intervention phases: elementary (grades 1–5) and secondary (grades 6–10) school. By young adulthood (ages 19–25), intervention participants reported lower weekly use of opioids in the past 12 months compared to control participants (1.7 % vs 4.1 %). The intervention was associated with a 61 % reduction in the likelihood of opioid use.

4. Discussion

Interventions targeting SDOH to improve OUD outcomes remain limited. This review identified 27 studies that sufficiently met criteria for SDOH-focused interventions and opioid use; this small number of eligible studies highlight the need for additional research in this area. Our review adopted a broad definition of SDOH interventions, recognizing that SDOH interventions may operate at multiple levels – from structural- and community-level interventions to individual-level efforts that address the consequences of social inequities. This inclusive approach allowed us to capture the current landscape of SDOH interventions but also revealed a critical gap: There is a scarcity of upstream interventions targeting the root causes of SDOH-related variations in terms of access to healthcare, OUD treatment, and supportive recovery services. Given that SDOH are known to influence the development, persistence, and outcomes of OUD and drive disparities in OUD outcomes (Lin et al., 2024), our findings substantiate areas that represent missed opportunities to intercede to curtail OUD and reduce health impacts and social consequences. Expanding interventions across multiple SDOH domains, as several included studies attempted to some degree, appears to be a viable approach to reduce disparities and improve OUD outcomes, including among individuals with SDOH factors that limit their access to needed services. Targeted interventions can improve health outcomes and achieve long-term societal savings (Thornton et al., 2016).

This scoping review revealed several key gaps in this area of research. Among the reviewed articles, there was a lack of breadth within each SDOH domain. Most (52 %) focused on the healthcare domain, aiming to improve provider knowledge, practice, and quality, leading to reduced opioid use (Alexandridis et al., 2019; Deshazer et al., 2020; Hershberger et al., 2022; Howard and Clark, 2017), or improved treatment initiation (Crowthers et al., 2022; Gryczynski et al., 2021; Hyde et al., 2022; Whiteside et al., 2021). Few studies addressed other factors, like OUD stigma among providers (Hershberger et al., 2022; Howard and Clark, 2017), linguistic and cultural competency, or insurance coverage, indicating missed opportunities (National Quality Forum, 2022). In the neighborhood and environment domain, most interventions relied on Housing First components to reduce opioid use (Crowthers et al., 2022; Dahlem et al., 2021; Davidson et al., 2014; Kelleher et al., 2021; Ly et al., 2024; Tuten et al., 2018), but none addressed other factors like the built environment outside of participants' own housing or neighborhood safety. Articles addressing the domains of neighborhood and economic stability often focused on recovery navigator interventions linking individuals to housing or employment support, which was associated with reduced opioid use (Crowthers et al., 2022; Holtyn et al., 2014; Novak et al., 2022). Notably, Kelleher et al. (2021) provided rental assistance (neighborhood and environment domain) and did not find any opioid use among at-risk participants within six months of the project, while Yoo et al. (2022) offered cash provisions (economic stability domain), which did not increase use.

Community-wide interventions influenced social supports and contexts, which were associated with increased abstinence (Alexandridis et al., 2019; Crowthers et al., 2022; Dahlem et al., 2021), while family-context interventions reduced opioid use (Godwin et al., 2020; Schaeffer et al., 2021). Notably, recent studies have used coalitions or mass media campaigns to increase awareness of OUD (Palombi et al., 2019), decrease stigma of people with OUD (Davis et al., 2024), and improve access to OUD treatment (Alexandridis et al., 2019; Glasgow et al., 2024). Given the evidence linking social cohesion and community support to reduced OUD, there may be additional opportunities for interventions that address community-level factors like stress, social cohesion, or civic participation (Avery et al., 2021; Dasgupta et al., 2018; Hser, 2007; Warren et al., 2007). Interventions in the education domain were understudied, with one focusing on elementary and secondary education to prevent opioid use in adulthood (Godwin et al., 2020), and no interventions addressing other education factors, such as high school graduation rates or education quality, highlighting a gap in OUD prevention strategies (Braveman, 2023; Lin, 2024). Overall, the limited focus on broader SDOH factors such as neighborhood, community, economic, and educational factors combined with the interrelated and interacting nature of these domains, underscoring the need for comprehensive, multi-faceted interventions to effectively address the spectrum of SDOH influence on OUD (ODPHP, 2024).

SDOH interventions were implemented in a limited number of settings but spanned the United States. Multiple studies were conducted in Massachusetts, New York, North Carolina, Maryland, and Ohio, representing both urban (e.g., Baltimore) and rural areas (e.g., rural northeast Minnesota). However, most interventions were conducted in healthcare settings (e.g., hospitals or SUD treatment centers), and fewer were conducted within community settings despite the importance of SDOH interventions designed with input

from people from the surrounding community. Community settings play a critical role in addressing the broader social and environmental factors influencing OUD outcomes. Interventions that are conducted in community settings, that address the community and social context, and that include community members (e. g., people with lived or living experience) can empower local communities to develop tailored solutions for reducing OUD risk and improving OUD outcomes (Bermea et al., 2019; Collins et al., 2019; Feld et al., 2023; Palombi et al., 2019; Rhodes et al., 2005; Singer, 2000). This empowerment is exemplified in the study by Palombi and colleagues, which showed a promising first step toward community-identified solutions (2019).

SDOH interventions have methodological and design issues that make it challenging to evaluate their effectiveness. Firstly, inconsistent terminology limits SDOH intervention evaluations. Equating SDOH to individual-level social risk factors further complicates the ability to effectively evaluate SDOH interventions that operate at structural and system levels (Alderwick and Gottlieb, 2019; Braveman and Gottlieb, 2014). Secondly, evaluating impacts of SDOH interventions has been limited by variations in study design. Over a third of studies used one-group quasi-experimental designs. These designs can be useful when randomization is not feasible, but they have limited protection against threats to internal validity and are more susceptible to biases. Although more than half of the articles on SDOH interventions included RCTs, which are considered the gold standard for establishing effectiveness, many studies focused on feasibility and acceptability rather than more meaningful efficacy outcomes (Dahlem et al., 2021; Smith et al., 2022). Utilizing RCTs may be challenging for upstream SDOH interventions due to the complexity of causal pathways, the long periods required to observe outcomes, variability in population responses, ethical dilemmas around randomization, and a perceived lack of methodological alternatives, all of which can limit the assessment of upstream intervention effects (Braveman, 2023). Moreover, many SDOH interventions incorporate several components to address multiple SDOH domains or social needs. For example, studies utilized peer navigators to connect patients to a variety of social services, including finding and maintaining employment or making social connections (Crowthers et al., 2022; Dahlem et al., 2021; Gryczynski et al., 2021; Tuten et al., 2018). Other interventions addressed multiple social needs such as housing, social support, education, or transportation provisions as part of service navigation or disease management strategies (Gryczynski et al., 2021; Whiteside et al., 2021), while housing supports were often embedded within holistic interventions (Crowthers et al., 2022; Dahlem et al., 2021), making it difficult to isolate the impact of specific intervention components. To address these challenges, researchers should adopt standardized definitions and frameworks for SDOH that include both individual and structural factors (Andermann, 2018) and clearly distinguish between social needs, which are often direct impacts of SDOH, and structural-level SDOH, which encompass the broader contextual factors influencing health outcomes (Castrucci, Auerbach, n.d.). Effective measurement strategies could include standardized tools capturing downstream SDOH contexts or social needs, such as housing stability, income, education, and healthcare access (Khurana et al., 2022). Future research should prioritize the development and validation of comprehensive, multi-component evaluation frameworks that can assess the complex and multifaceted nature

of SDOH interventions, as well as the complexities of the SDOH domains they intend to address.

This review is among the first to evaluate SDOH interventions for their ability to improve OUD outcomes in the U.S. We conducted a comprehensive review of the current literature using multiple search strategies and two independent reviewers to minimize bias, with a pre-established review protocol for transparency. Despite these strengths, the study has limitations. First, the search was limited to articles published within an 11-year period in the U.S. (from 2013 to 2024). For example, this review may have missed SDOH-informed prevention strategies because they were published prior to our inclusion window or did not include the word “opioid” within the abstract. Future research should include a focused scoping review of SDOH-informed prevention strategies aimed at reducing later opioid use. Such a review may reveal additional family-oriented or school-based interventions (Spoth et al., 2008). Second, there is no standardized MeSH term for SDOH, which may lead to missing papers. Third, the study design criteria for inclusion may have resulted in fewer identified SDOH interventions, but it ensured that only studies assessing intervention effects were included. Fourth, although we conducted a hand search of Google Scholar to capture gray literature, this review did not identify any gray literature. Finally, some studies were excluded because they were SDOH-adjacent or did not meet study design requirements (e.g., comparison group or time point). For example, we excluded studies focusing on patient health literacy, knowledge, and problem-solving (Braveman, 2023). We defined education as education attainment and therefore excluded interventions that educated patients about OUD or OUD treatments (H. E. Jones et al., 2021; J. D. Jones et al., 2022; Kelleher et al., 2021; LaSane et al., 2022; Lott and Rhodes, 2016; McElhinny et al., 2021; Morasco et al., 2022). However, some studies that were excluded due to specific design issues that impacted reporting of their results still deserve mention. For instance, The Safe Passages program allowed people with OUD to enter the police department to seek treatment without fear of arrest (social and community context domains), but relied on qualitative methods (Jurgens et al., 2024). The MAT-PLUS intervention created a supportive family context to improve adherence with MOUD as long-acting injectable naltrexone (social and community context domains) but the article did not report OUD results separately (Wenzel et al., 2021), nor did the START study (Ober et al., 2023), which provided addiction consultation and support services to hospitalized patients, aiming to enhance healthcare quality (healthcare domain). Despite these limitations, this review provides valuable insights into SDOH interventions to improve OUD outcomes and highlights areas of opportunity needing additional research. Finally, while our review focused on interventions addressing SDOH domains, many studies used downstream and midstream interventions.

5. Conclusions

There is a pressing need to understand the potential impact of SDOH interventions on OUD outcomes. The majority of SDOH interventions we identified were focused on individual-level approaches to mitigate the effects of structural inequities. This reflects a broader trend in the field and underscores the need for more upstream interventions. Thus, our findings underscore public health calls for interventions to address root causes (Frieden, 2010). Policies to improve population-level wages, enhance healthcare systems,

and provide economic support and educational opportunities may have the most significant impact on population-level health, yet are challenging to implement (Frieden, 2010). SDOH interventions, including cash assistance, vocational training, support for education attainment, social reentry programs, and supportive social and physical environments, may mitigate factors associated with OUD and may be easier to design and implement than population-level interventions. Public health interventions that integrate both upstream and downstream strategies are essential to improving the lives of people with OUD. To maximize the effectiveness of these strategies, however, it is essential to resolve measurement issues to best understand the impacts of these approaches. Harmonizing SDOH terms and metrics across studies, along with developing standardized measurement tools and outcome measures specific to SDOH domains, is crucial for advancing research and comparing study results.

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

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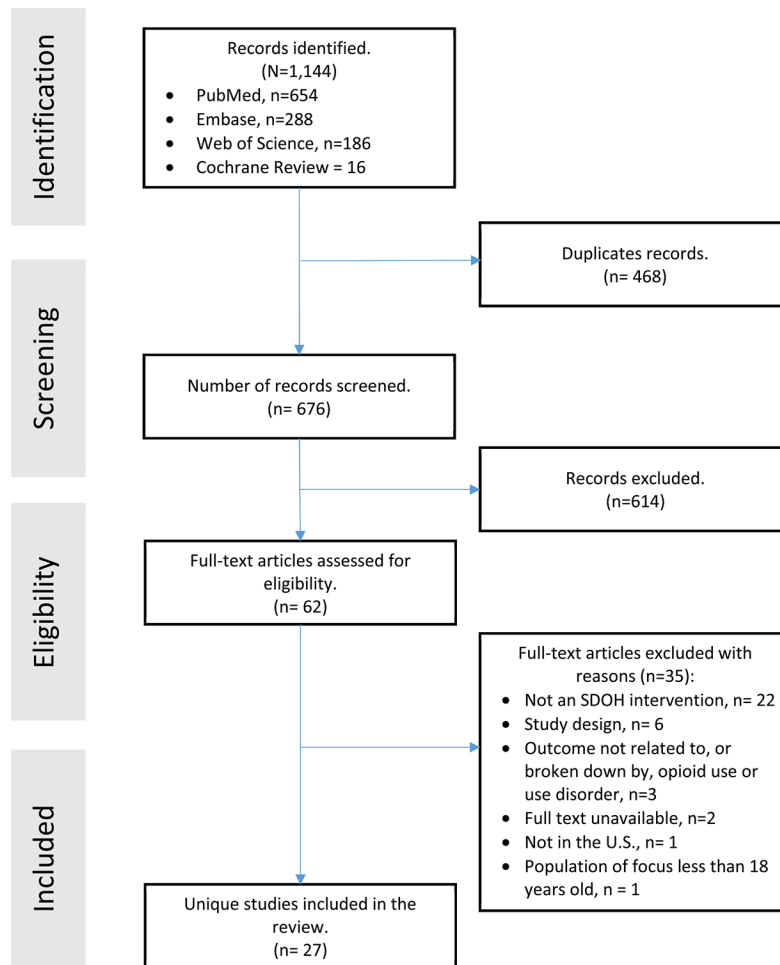


Fig. 1.
Flowchart Outlining Search Strategy.

Table 1

Example of SDOH Conditions and OUD Risk.

Domain	Domain Examples	OUD Examples	Examples from the Literature
Healthcare system	Quality of care, provider availability, linguistic and cultural competency, and healthcare coverage.	quality of OUD treatment, provider negative bias about people with OUD, access to evidence-based OUD treatments, access to medication for opioid use disorder, systematic and consistent referrals to appropriate levels of care.	Medications for OUD are likely in treatment programs that have greater medical resources and funding (Abraham et al., 2020; Knudsen and Roman, 2014).
Education	Education attainment, early child education, vocational training, and higher education	early childhood education, vocational supports, access to or achievement of higher education	Adults who did not attend college have disproportionately high rates of substance use disorders (Braveman, 2023).
Community and social context	Social integration, support systems, community engagement, discrimination, and stress.	Family, neighborhood, mutual aid and other social support systems, engagement or reentry social supports, discrimination, stress, isolation, stigma.	Social support as well as living in communities with social cohesion and community beneficence is associated with lower rates of OUD (Avery et al., 2021; Dasgupta et al., 2018; Hser, 2007; Warren et al., 2007).
Economic Stability	Employment, income, expenses, debt, medical bills, and economic support.	Supports for supplemental income, expenses, debt, medical bills, and other economic supports.	Adults who are unemployed have disproportionately high rates of substance use disorders (Braveman, 2023) whereas employment is associated with OUD recovery (Hser et al., 2015)
Neighborhood and physical environment	Housing, transportation, safety, access to playgrounds, parks, walkability and zip code or geographic location.	Transportation options, availability and cost of housing, safety/violence, geographic setting (i.e., rurality).	Risk of fentanyl-involved overdose death varies by neighborhood disadvantage (Wagner et al., 2021). Patients living in rural areas vs. urban residents have greater challenges in accessing OUD treatment due to lower specialized treatment options, longer travel time to treatment facilities, higher economic burden, and less health insurance (Bond Edmond et al., 2015; Kiang et al., 2021).

Domain examples column are from Kaiser Family Foundation. (2018). Beyond health care: The role of social determinants in promoting health and health equity. <https://www.kff.org/racial-equity-and-health-policy/issue-brief/beyond-health-care-the-role-of-social-determinants-in-promoting-health-and-health-equity/>

Table 2

Characteristics of Included Studies.

	N (%)
SDOH Domains	
Healthcare system	16 (37 %)
Community and social context	13 (30 %)
Economic Stability	7 (16 %)
Neighborhood and physical environment	4 (9 %)
Education	1 (2 %)
Settings	
Healthcare settings	14 (52 %)
Community settings	12 (44 %)
School settings	1 (4 %)
Populations	
People in jeopardy of developing OUD	8 (21 %)
People with OUD	22 (60 %)
Providers or other staff	7 (18 %)
Study Design	
Randomized Controlled Trial	14 (52 %)
Quasi-experimental design two group design	3 (11 %)
Quasi-experimental one group, post-test only, etc.	10 (37 %)

Note: Articles covered multiple domains of SDOH (n = 41) and multiple populations (n = 37) therefore do not add up to 27.

Table 3

Study Design by SDOH Domains.

	N (%)
SDOH Domains	
Healthcare system (n = 16)	
Randomized controlled trial	6 (38 %)
Quasi-experimental (two-group design)	2 (12 %)
Quasi-experimental (one-group design)	8 (50 %)
Community and social context (n = 13)	
Randomized controlled trial	3 (23 %)
Multi-site, community-level, cluster-randomized trial	2 (15 %)
Quasi-experimental (two-group design)	1 (8 %)
Quasi-experimental (one-group design)	7 (54 %)
Economic Stability (n = 7)	
Randomized controlled trial	4 (57 %)
Quasi-experimental (one-group design)	3 (43 %)
Neighborhood and physical environment (n = 4)	
Quasi-experimental (one-group design)	4 (100 %)
Education (n = 1)	
Randomized controlled trial	1 (100 %)

Note: Articles covered multiple domains of SDOH (n = 41) therefore the overall design types do not add up to 27.

Table 4

Summary of Eligible Articles (N = 27).

Article	Design	Intervention Strategies	SDOH Domains	Population	Setting	Location	Outcomes	Main findings
Alexandridis et al., (2019)	Quasi-experimental. Interrupted time series comparing before and after implementation (one group design)	Community support systems Healthcare-provider training, expansion of addiction treatment Emergency department policy revision to limit opioid analgesics prescription	Community and social context; Healthcare system	Community general population Buprenorphine patients Prescribers within 100 counties over 2009–2014 (N = 7200 county-months)	74 counties	North Carolina	Not reduced prescribing of opioid analgesics. Not decreased buprenorphine utilization.	Implementation of Project Lazarus (PL), a seven-strategy, community-coalition-based intervention, did not appreciably reduce opioid dispensing or increase buprenorphine utilization.
Crowthers et al., (2022)	Quasi-experimental. One group pretest posttest.	Peer recovery coaching which covered living and financial independence, employment and education, relationships and social support, and independence from legal involvement and institutions	Economic stability; Neighborhood/physical environment; Community and social context; Healthcare system	Male patients with SUD and/or co-occurring mental health disorder (N = 117).	Maryville Addiction Treatment Centers	New Jersey	A higher rate of abstinence. Increased housing stability. Lower mental health and behavioral/social consequences. Longer participant-recovery coach exposure time and higher follow-up rates.	Patients paired with a recovery coach extensively trained in MOUD can achieve a variety of desired outcomes such as higher drug abstinence, increased housing stability, improved treatment retention, and reduced negative health and social consequences.
Dahlem et al., (2021)	Quasi-experimental. One group pretest posttest. Pilot intervention.	Peer recovery coach supporting housing, transportation, clothing, insurance, food stamps, household items, phone, identification cards, insurance, and referral to treatment services	Economic stability; Neighborhood and physical environment, Food, Community and social context, Healthcare system	Patients who experienced non-fatal overdose (N = 122).	Community setting involving local crisis team, law enforcement agencies, SUD/mental health treatment facilities, and ED	Washtenaw County, Michigan	Among 122 overdose survivors, 77.0 % were engaged, 33.6 % (n = 41) received ongoing treatment services, one participant received recovery housing, three (2.5 %) were incarcerated and three (2.5 %) were deceased, and 75 (61.5 %) declined services.	The community-wide coordinated program in the EDs showed promise in linking overdose survivors to recovery support and treatment services post-overdose.
Davidson et al., (2014)	Quasi-experimental. One group pretest posttest. Feasibility study.	Supportive housing using participant involvement (“consumer-participation”) for chronically homeless individuals whose substance use is a primary barrier to independent housing. Providers were expected to follow a	Neighborhood and physical environment	Individuals with histories of chronic homelessness and problematic substance use (N = 358).	Community setting: 500 units of apartments	New York City, New York	Fidelity in implementation of supportive housing components and client substance use was not related. However, clients in participant involvement (or “consumer participation-consistent programs”) were less likely than others to report using	Clients in programs with greater fidelity to participant involvement components of Housing First were more likely to be retained in housing and were less likely to report using stimulants or opiates at follow-up.

Article	Design	Intervention Strategies	SDOH Domains	Population	Setting	Location	Outcomes	Main findings
		Housing First approach for providing client-centered supportive services to improve retention.					stimulants or opiates at follow-up.	
Davis et al., (2024)	Multi-site, parallel-group, cluster-randomized, waitlist-controlled trial	Utilized a Communities that HEAL (CTH) intervention on perceived opioid-related community stigma by stakeholders in the HEALing Communities Study (HCS). Communities selected CTH interventions from a menu of options based on their existing resources, needs, and service gaps, thus, EBP's varied by community	Community and social context	Adult community coalition members and key stakeholders in the 66 Intervention and Wait-list Control HCS (N = 1385). There were 1385 coalition stakeholders in the sample, representing all 66 participating communities. The substance use sector represented 47 % of all sectors.	Community setting	67 communities in four states—Massachusetts, New York, Ohio), and Kentucky (	Intervention stakeholders reported a larger decrease in perceived community stigma toward people treated for OUD (adjusted mean change (AMC) -3.20 [95% C.I. -4.43, -1.98]) and toward MOUD (AMC -0.33 [95% C.I. -0.56, -0.09]) than stakeholders in Wait-list Control communities (AMC -0.18 [95 % C.I. -1.38, 1.02], p = 0.0007 and AMC0.11 [95% C.I. -0.09, 0.31], p = 0.0066). The relationship between intervention status and change in stigma toward MOUD was moderated by rural-urban status (urban AMC -0.59 [95 % CI, -0.87, -0.32], rural AMC not sig.) and state.	The CTH intervention decreased stakeholder perceptions of community stigma toward people treated for OUD and stigma toward MOUD. Implementing CTH intervention in other communities could decrease OUD stigma across diverse settings nationally
Deshazer et al., (2020)	Quasi-experimental retrospective study with pre-post intervention study	To reduce access to opioids the study conducted Quality management-focused reducing over prescribing of opioids. For those circumstances where opioids were indicated, implement tools and controls to encourage safe prescribing of opioids consistent with the CDC guidelines and de-escalating protocols for those on high doses (secondary prevention). Among those with OUD, offered interventions	Healthcare system	Records from OUD diagnosed commercial members per year (N = 9700).	Healthcare (national health plan) and community setting involving local, state, and federal agencies; local school districts; businesses; law and drug enforcement; and community-and faith-based organizations	Delaware, Pennsylvania, West Virginia	A 19 % decrease across the eligible Highmark membership of total opioid prescription fills normalized per 1000 members. A 15 % decrease was observed between 2016 and 2017, and this was a sharper decrease than observed in previous years between 2013 and 2016. In terms of RX fills by geography, the results varied by state between 2016 and 2017, with the biggest decreases in WV and PA at 16 % each and 10 % in	Reduced total Opioid RX fills, opioid use by dose and duration for members diagnosed with OUD. A decrease of 19 % of total Opioid RX fills; shorter durations and the majority of these members switched to 7 days or less of opioid use; and a reduction by 13 percentage point of the number of members on higher strength 20 + MMEs opioids resulted.

Article	Design	Intervention Strategies	SDOH Domains	Population	Setting	Location	Outcomes	Main findings
		to support a lasting recovery (Tertiary prevention)						
Glasgow, et al., (2024)	Multi-site, parallel-group, cluster-randomized, waitlist-controlled trial	The Communities That HEAL (CTH) intervention, a multi-site, community-level, cluster randomized waitlist controlled trial, seeks to reduce opioid overdose deaths through evidence-based practices, community engagement, and health communication campaigns in 67 highly impacted communities to increase access to behavioral health access	Community and social context	67 communities in four states	Community setting	67 communities in four states—Massachusetts, New York, Ohio), and Kentucky (	No significant differences were detected between intervention and waitlisted communities in the rate of individuals receiving any of the five BHS categories. None of the interaction effects used to test the effect modification were significant.	No significant intervention effects were observed through Medicaid claims data, the best available data source but limited in terms of capturing individuals reached by the intervention. Also, the 12-month evaluation window may have been too brief to see improved outcomes considering the time required to stand-up BHS.
Godwin and Conduct Problems Prevention Research Group, (2020)	Randomized controlled trial	Fast Track (FT), a comprehensive childhood intervention designed to decrease aggression and delinquency in at-risk kindergartener by targeting children's intrapersonal, interpersonal, and academic skills and their parents' parenting skills and behaviors through two intervention phases: elementary (grades 1–5) and secondary (grades 6–10) school.	Education, Community and Social Connectedness	Participants (N = 713) from high-risk elementary schools (N = 55) based on neighborhood level crime and poverty.	School setting involving elementary schools and parent-child relationships	Durham, North Carolina; Nashville, Tennessee; rural Pennsylvania; and Seattle, Washington	During young adulthood (ages 19–25), control participants more often reported weekly use of opioids in the past 12 months compared to intervention participants. Intervention significantly decreased the probability of opioid use by 61.2 %. Control participants more often reported hazardous drinking between ages 15 and 25, compared to intervention youth. Intervention significantly decreased the probability of hazardous drinking by 45.3 %.	Intervention improvements to interpersonal skills (e.g., prosocial behavior, authority acceptance) are the strongest indirect pathway to reduce harmful behaviors (despair, suicide ideation, self-harm, hazardous drinking, opioid use).
Gryczynski et al., (2021)	Two group parallel randomized trial	Intervention group consisted of patient navigators to conduct proactive case management, advocacy, service linkage, and motivational support to resolve internal and external barriers to care and address medical and basic needs (such as	Economic stability, Healthcare system	Hospitalized adults with comorbid SUDs of opioid, cocaine, or alcohol (N = 400).	Healthcare - A large urban academic medical hospital with an SUD consultation service	Baltimore, Maryland	Participants in the navigator intervention group were less likely to have an inpatient readmission within 30 days compared to participants receiving TAU and were more likely to enter community SUD treatment after discharge/	Patient navigation reduced inpatient readmissions and ED visits in this clinically challenging sample of hospitalized patients with comorbid SUDs.

Article	Design	Intervention Strategies	SDOH Domains	Population	Setting	Location	Outcomes	Main findings
Hagedorn et al., (2022)	Quasi-experimental. Two group pre/post test with block randomization.	Healthcare - Quality improvement - assessment of local barriers/facilitators, formation of a local implementation team, a site visit for action planning and training/education, cross-facility quarterly calls, monthly coaching calls, and consultation.	Healthcare system	VHA facilities (N = 35).	Healthcare - Veteran Health Administration facilities in the lowest quartile of performance on the MOUD/OD ratio among all VHA facilities as of October 2017.	Unknown	Intervention facilities significantly increased the ratio of patients with OD receiving MOUD from an average of 18 % at baseline to 30 % 1 year later, with an absolute difference of 12 %.	Intensive external facilitation improved the adoption of MOUD in most low-performing facilities and may enhance adoption beyond other interventions less tailored to individual facility contexts.
Hershberger et al., (2022)	Randomized control trial	Healthcare quality - Online training simulation of a healthcare encounter utilizing virtual reality immersion	Healthcare system	Clinical and nonclinical health professionals (N = 158).	Health organizations, such as community health centers	Unknown	Among intervention participants, there was an increased level of compassion for the simulated character who was pregnant woman and using opioids.	Simulation online trainings can heighten awareness in the increase of increasing the cultural sensitivity of clinicians in health care professions for improving health equity.
Hollyn et al., (2014)	Randomized control trial	Economic stability - Assigned to one of three treatments: Work Reinforcement (work to earn pay), Methadone & Work Reinforcement (enroll in methadone treatment to work and maximize pay), and Abstinence, Methadone & Work Reinforcement conditions (provide opiate-and cocaine-negative urine samples to maximize pay).	Economic stability	Unemployed adults with OD who use injection drugs and not currently in treatment (N = 98).	Therapeutic workplace at the Johns Hopkins Bayview Medical Center at large hospital	Baltimore, Maryland	Increased enrollment in methadone treatment during induction. Drug abstinence increased as a graded function of the addition of the methadone and abstinence contingencies. Abstinence, Methadone & Work Reinforcement participants more often provided urine samples negative for opiates and cocaine compared to Work Reinforcement participants. Methadone & Work Reinforcement participants provided significantly more cocaine-negative samples than Work Reinforcement participants.	The therapeutic workplace can promote drug abstinence in people who inject drugs and are out-of-treatment.
Howard and Clark, (2017)	Quasi-experimental. One group pretest posttest.	Educational intervention for health professionals based on an interprofessional-shared	Healthcare system	The purposeful sample (N = 45) consisted of health care providers in the	The purposeful sample (N = 45) consisted of health care providers in	Rhode Island	Knowledge about child welfare laws and resources, stigma reduction, and provider	Increased their knowledge of child welfare laws pertaining to prenatal substance

Article	Design	Intervention Strategies	SDOH Domains	Population	Setting	Location	Outcomes	Main findings
		decision making (IP-SDM) model.		obstetrical setting. There was a total of 45 maternal health care providers who received the training.	the obstetrical setting. There was a total of 45 maternal health care providers who received the training.		referrals to the RI Department of Health's Healthy Families America Program (HFA).	use as well as what community resources might be available to this population of women. Increased referrals to an evidence-based program, Healthy Families America. Stigma was reduced from pre to post training regarding women with substance use disorders.
Hyde et al., (2022)	Quasi-experimental prospective cohort design with an historical comparison group.	Post-Incarceration Engagement (PIE) intervention is a peer-based enhancement designed for the VA healthcare for reentry veterans program. Social and emotional support, linkage and referral to healthcare and social services, and role modeling of life skills.	Community and social connectedness. Healthcare system/Access	Veterans postincarceration. PIE group (n = 43) and historical comparison (n = 36).	Veterans postincarceration. PIE group (N = 43) and historical comparison (N = 36).	Massachusetts	No difference in linkage to primary care within 90 days of release, (58 % versus 67 %). Intervention participants were significantly more likely to receive substance use treatment than the comparison group (86 % versus 19 %, $p < .0001$) and the mean monthly substance use visits was greater in the intervention group (0.96 versus 0.34, $p < .007$). Engagement in mental health services was greater for the intervention group than the comparison group (93 % versus 64 %, $p < .003$). There were no significant differences between groups for emergency department use and hospitalization.	Intervention was found to be feasible and contributed to returning citizens' greater linkage and engagement with substance use, mental health, and other specialty care than a historical comparison group.
Kelleher et al., (2021)	Quasi-experimental. One group pretest posttest. Feasibility study.	Housing first: 6-month rent plus utilities and Strength-Based Outreach and Advocacy and prevention interventions focused on HIV and opioids.	Neighborhood-Housing	18-24 years who were lacking a fixed, regular, stable, and adequate nighttime residence (N = 21).	Community in a Large Midwestern City-recruitment was at a drop-in center for homeless youth.	Columbus, Ohio	A total of 17 youth completed the study (85 % retention), and a high proportion of youth were stably housed at 6-month follow-up. Participation in intervention services was high with an average of 13.57 sessions for advocacy, 1.33 for MI, and 0.76 for HIV prevention. Alcohol	The Housing First model appeared to be feasible to deliver, and youth engaged in the supportive intervention services. The study demonstrates the potential for an adapted Housing First model to be delivered to youth experiencing homelessness and may improve outcomes,

Article	Design	Intervention Strategies	SDOH Domains	Population	Setting	Location	Outcomes	Main findings
Ly et al., (2024)	Quasi-experimental one group pretest posttest. Pilot study.	Pharmacist-led pilot service in providing more equitable and accessible BUP treatment to individuals residing in Permeant Supporting Housing	Healthcare system	Formally unhoused participants who lived in Permeant Supportive Housing (N = 38)	Permeant Supportive Housing with SUD	California	use did not change significantly over time. However, drug use, drug use consequences, and cognitive distortions, and the size of youths' social networks that were drug using individuals decreased significantly.	A pharmacist-led BUP outreach service was found to increase treatment adherence in individuals residing in PSH over 3 months. Low-barrier BUP treatment models, such as that evaluated in this study, may help provide more equitable and
McCarthy et al., (2019)	Randomized controlled trial. Three-arm prospective, randomized controlled pragmatic trial with randomization occurring at the physician level.	Enhanced provider-patient communication through an Electronic Medication Complete Communication (EMC2) Opioid Strategy text reminders to patients to ED patients about newly prescribed opioid pain relievers. Alerts and messages were sent to ED physicians, primary care providers, and pharmacists to counsel patients on safe opioid use. Patients also received text messages on safe opioid use.	Healthcare system	Patients being discharged from an urban academic ED with a new prescription for hydrocodone-acetaminophen (N = 652).	Urban, academic emergency department	Chicago, Illinois	Demonstrated safe opioid use occurred more often in the EMC2 group (adjusted odds ratio [aOR] = 2.46, 95 % confidence interval [CI] = 1.19-5.06), but not the EMC2 + SMS group (aOR = 1.87, 95% CI = 0.90-3.90) compared with usual care.	The study found that the intervention improved increased likelihood of safe use of opioids compared to TAU and increased patient knowledge. However, there was no influence of the intervention on safe medication use among the portion of the sample returning medication diaries.
Molfenter et al., (2019)	A cluster randomized control trial	A multi-organizational learning collaborative to reduce barriers to reimbursement and access to buprenorphine. The intervention included three learning sessions at the beginning, mid-	Healthcare system	Patients with OUD diagnosis (N = 22295).	Forty-eight of the 53 eligible treatment agencies in the 14 ADAMHS board areas enrolled in the trial.	Ohio	The intervention condition experienced a 3.0-fold increase in the percentage of buprenorphine use by OUD patients during the experimental period (9.5 % baseline; 28.5 % experimental).	The payer-provider partnering process increased payers' ability to request greater use of buprenorphine. Partnering also led to payers helping link providers to financial

Article	Design	Intervention Strategies	SDOH Domains	Population	Setting	Location	Outcomes	Main findings
		point, and end of a 24-month period, with monthly coaching calls in between. Sessions and calls were directed at payers and treatment agencies to support organizational changes and increase buprenorphine access. Used policy levers to activate systems change, strategizing how payers and providers could overcome barriers to treatment access.					Comparison ADAMHS boards and their providers experienced a significantly lower increase in percentage of buprenorphine use (10.9% baseline; 19.6% experimental) with an adjusted difference-in-differences of 10.3% (95% CI [9.9%, 10.7%]), $p < .001$).	and non-financial resources needed to remove barriers and develop buprenorphine treatment infrastructure. Coaching technical assistance during phone calls assisted the payers in creating an environment for greater buprenorphine utilization as a percentage of OUD patients.
Novak et al., (2022)	Randomized controlled trial	Participants who qualified for Phase 2 were randomly assigned to either the usual care control group or the abstinence-contingent wage supplement group.	Economic-income	Participants were unemployed adults in opioid agonist treatment (N = 91)	Opioid agonist treatment	Baltimore, Maryland	Abstinence-contingent wage supplement participants provided significantly more opiate- and cocaine-negative urine samples than usual care control participants (63.6% vs. 44.1%); abstinence-contingent wage supplement participants were also significantly more likely to become employed and live out of poverty than usual care participants during intervention.	Long-term delivery of abstinence-contingent wage supplements can promote drug abstinence and employment, but many patients relapse to drug use and cease employment when wage supplements are discontinued.
Palombi et al., (2019)	Quasi-experimental. One group pretest posttest. Feasibility and effectiveness study.	Collaborative teams planned and implemented a series of nine community forums focused on opioid and heroin use across rural northeast Minnesota to educate and unite invested community members on the critical public health issue.	Community and social context	Rural community forums completed surveys to assess measures of growth in knowledge and awareness (N = note stated)	Community forums	Rural northeast Minnesota	Collaborative teams planned and implemented a series of nine community forums focused on opioid and heroin use across rural northeast Minnesota to educate and unite invested community members on the critical public health issue.	Community forums have functioned as an effective grassroots approach to engaging rural community members in opioid use prevention and intervention efforts.
Quanbeck et al., (2018)	A randomized controlled trial with matched-pair design. Feasibility, acceptability and	'Systems consultation' strategy with audit and feedback, academic detailing (face-to-face education), and external facilitation (process management	Healthcare system	Community-based primary care clinics (N = 8)	Primary care clinics	Wisconsin	At 6 months, statistically significant improvements were noted in intervention clinics in the percentage of patients with mental health screens,	The systems consultation implementation strategy demonstrated feasibility, acceptability, and effectiveness in a

Article	Design	Intervention Strategies	SDOH Domains	Population	Setting	Location	Outcomes	Main findings
	effectiveness study.	support) to enhance adherence to opioid prescribing guidelines in primary care. Promoted organizational changes to improve care quality through tailored support and systems engineering tools, translating guidelines into a checklist with expert input, forming change teams, and supporting implementation with monthly site visits and teleconference					treatment agreements, urine drug tests, and opioid-benzodiazepine coprescribing. At 12 months, morphine-equivalent daily dose was significantly reduced in intervention clinics compared to controls.	study involving eight primary care clinics. This multidisciplinary strategy holds potential to mitigate the prevalence of opioid addiction and ultimately may help to improve implementation of clinical guidelines across healthcare.
Schaeffer et al., (2021)	Randomized controlled trial	"Multisystemic Therapy – Building Stronger Families integrates Multisystemic Therapy for Child Abuse and Neglect" (MST-CAN) and Reinforcement Based Treatment (RBT) as a family-centered intervention delivered by community-based therapists. Intervened in the home, community, and school settings, aiming to improve family dynamics and the broader social environment through parenting skills training and cognitive behavioral therapy.	Community and social context	Families who had an open case with Child Protective Services (N = 98).	Community	Two areas of the state of Connecticut	Parents reported significant declines over time in the number of days they used in the past month. Parents on average were increasingly more likely to report total past-month abstinence from all drugs (p = .002), from alcohol (p = .001), and from cocaine (p = .045). Parents who received MST-BSF were marginally more likely to report total abstinence from opioids (p = .055) and from cocaine (p = .055).	The findings of this study add to previous pilot research supporting the effectiveness of MST-BSF, delivered by community-based therapists, to significantly reduce parental substance misuse and child neglect through a family and ecologically based treatment.
Shaffer et al., (2021)	Quasi-experimental. One group pretest posttest. Open pilot design.	MISSION-CJ treatment for justice-involved individuals with COD, including the integration of criminogenic risk and needs assessment, risk-need-responsivity (RNR) informed treatment planning, and materials that promote prosocial thinking and behavior.	Community and Court-justice	Patients in Drug Treatment Courts (DTCs), N = 79	2 DTCs	Urban area in Massachusetts	Reductions in illicit substance use. Regarding opioid use, there was a significant effect of time in MISSION-CJ in reducing opioid use (p = 0.0013), and reductions were observed for clients with high engagement in MISSION-CJ unstructured sessions (p = 0.0387).	This pilot study demonstrated that MISSION-CJ could be implemented alongside DTC for a relapsing DTC population with COD to enhance criminal justice and behavioral health outcomes. This study observed high treatment engagement, from baseline to 6-month follow up.

Article	Design	Intervention Strategies	SDOH Domains	Population	Setting	Location	Outcomes	Main findings
Suzuki et al., (2023)	Randomized controlled trial	Utilized peer recovery coaches, to intervene with hospitalized patients with OUD, providing a relapse prevention plan, contact for six months post-discharge contact, encouragement of MOUD adherence, and identifying community resources	Healthcare	Patients with OUD (N = 25)	Hospital	Massachusetts	No significant differences were found in the proportion of participants retained in MOUD treatment at 6 months (38.5 % vs 41.7%, P = 0.87), proportion of participants readmitted at 6 months (46.2 % vs 41.2 %, P = 0.82), or the time to treatment discontinuation (log-rank P = 0.92) or readmission (log-rank P = 0.85).	This pilot trial did not demonstrate that a recovery coach intervention improved MOUD treatment retention compared with treatment-as-usual among hospitalized individuals with OUD.
Tuten et al., (2018)	Quasi-experimental design with a comparison group (Two groups).	The intervention of recovery housing is to provide a communal, supervised living environment for individuals with substance use disorders (SUDs).	Community, economic stability Neighborhood and physical environment	Patients (N = 135) with OUD receiving reinforcement-based treatment with or without recovery housing	Community	Maryland	Participants who accessed recovery housing at any point during treatment were more likely to be opioid abstinent (at one, three and six months).	These analyses found no significant group differences on abstinence and employment outcomes, thus suggesting the potential efficacy of providing recovery house along with RBT.
Whiteside et al., (2021)	Randomized controlled trial. Pilot Study	Utilized an Emergency Department Longitudinal Integrated Care (ED-LINC) intervention included bedside brief negotiated interviews, overdose education, facilitation MOUD, proactive longitudinal care management, and the use of a health information exchange to improve healthcare access and continuity	Healthcare system	Participants at a large, urban, safety net hospital (N = 40)	Hospital emergency department (ED)	Seattle, Washington	At the end of the 3-month intervention, fewer days of heroin use in the past 30 days. No significant differences over the course of the 6-month investigation. No significant changes for any new MOUD initiation.	ED-LINC intervention did not result in an improvement in illicit opioid used and MOUD initiation.
Yoo et al., (2022)	Randomized control trial with parallel groups.	Randomly assigned to receive either a substantial (\$333) or a nominal (\$20) monthly cash gift during the early years of the infant's life.	Economic stability	Low-income mothers in the U.S. with newborns were recruited from hospitals shortly after the infant's birth (N = 1000)	Community	Nationwide	The cash gift difference of \$313 per month had small and statistically nonsignificant impacts on group differences in maternal reports of substance use and household expenditures on alcohol or cigarettes. The estimated share of the \$313 group difference spent on	Cash gifts do not impact opioid use.

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Article	Design	Intervention Strategies	SDOH Domains	Population	Setting	Location	Outcomes	Main findings
alcohol and tobacco was less than 1 %.								

Table 5

Description of Articles by Domains.

SDOH Domain	Focus	Author(s) & Year	Sample Size	Intervention	Key Results
Healthcare System	Provider knowledge and practice	Alexandridis et al., (2019)	4 counties	Multifaceted community-coalition-based intervention featuring healthcare-provider training and expansion of addiction treatment	n.s. reduction in prescribing of opioid analgesics n.s. decrease buprenorphine utilization
		Deshazer et al., (2020)	9700 patient records	Quality management-focused interventions	↓ prescription opioid fills by 19 % 84 % of members used opioids prescriptions for ≤ 7 days ↓13 percentage points in use of higher strength opioids ↑ knowledge of community supports by 38 %.
		Howard and Clark, (2017)	45 obstetrical health professionals	Training for obstetrical health professionals	↑ safe opioid use occurred more often in the consult group (aOR 2.46, CI 1.19, 5.06) and consult plus text reminders to patients (aOR 1.87, 0.90–3.90) vs. TAU
		McCarthy et al. (2019)	126 ED providers serving 652 patients	ED intervention to alert physicians to counsel patients on safe opioid use	80 % of staff endorsed being familiar with guidelines for safe opioid prescribing
	Quality assurance and treatment access	Quanbeck et al., (2018)	8 community-based primary care clinics	Quality improvement to implement clinical guidelines on opioid prescribing	n.s. difference in opioid use (IRR: 1.50) ↑ MOUD initiation (50 % vs. 30 %) ↑ rate of abstinence (X ² =20.0)
		Whiteside et al., (2021)	40 PWOOD	ED intervention with case management and education	33.6 % received ongoing treatment services
		Crowthers et al., (2022)	117 PWOOD	Peer recovery coaching featuring medical services and insurance services referrals	Faster entry into treatment ↑ SUD treatment entry within 3 months (50.3 % intervention vs 35.3 % TAU) Faster entry into treatment ↑ MOUD treatment entry within 3 months (68 % intervention vs 40 % TAU) ↓ positive opioid drug test within 6 months (68 % vs 74 % TAU)
		Dahlem et al., (2021)	122 patients who experienced non-fatal overdose	Community case management navigators and peer coaches	↑ patients receiving MOUD (18–30 % 1 year later, with an absolute difference of 12 % ↑ linkage and engagement in substance use (86 % vs. 19 %), mental health (93 % vs. 64 %)
		Gruczynski et al., (2021)	400 patients with co-occurring SUDs	Patient navigators in hospitals	↑ treatment adherence among intervention group (37 % post-intervention vs. 3 % pre-intervention, achieved 80 % of prescription days filled)
		Hagedorn et al., (2022)	35 facilities	Quality improvement interventions to remove barriers	
		Hyde et al., (2022)	43 previously incarcerated veterans	Post-incarceration peer-based intervention with linkage to care	
		Ly et al., (2024)	38 PWOOD	Pharmacist-led buprenorphine outreach services at Permanent Supportive Housing for people with OUD	

SDOH Domain	Focus	Author(s) & Year	Sample Size	Intervention	Key Results
Education	Stigma in healthcare settings	Molfenter et al., (2019)	48 facilities serving 22,295 patients with OUD	Quality improvement interventions with coaching calls to reduce barriers	↑ Threefold increase in patients receiving buprenorphine
		Shaffer et al., (2021)	79 individuals in Drug Court	Wrap-around services to establish links to behavioral health and other prosocial supports	↓ opioid use ↓ nights in jail
		Suzuki et al., (2023)	25 hospitalized patients with OUD	Recovery coaches initiated during hospitalization	n.s. MOUD retention vs. TAU (38.5 % vs 41.7 %, $p > .05$) n.s., readmitted (46.2 % vs 41.2 %, $p = 0.82$) n.s., time to treatment discontinuation (log-rank, $p = 0.92$) n.s., readmission (log-rank, $p = 0.85$)
		Howard and Clark, (2017)	45 providers	Training for obstetrical health professionals	↑ negative bias toward substance-using women by 73 % ↑ knowledge of community resources for OUD risk patients 7.8 % increase
Community and social context	Early education	Hershberger et al., (2022)	178 health professionals	VR training simulation	n.s. differences in overall bias scale ↓ frustration and ↑ compassion scores for pregnant women who use opioids (mean differences of $-.45$ and $-.31$). Weekly opioid use in past 12 months during ages 19–25: Control: 4.1 % Intervention: 1.7 % ↓ probability of opioid use by 61 %
		Godwin and Conduct Problems Prevention Research Group, (2020)	713 at-risk kindergartners	Social and emotional development, parenting, children's skills & behaviors	↑ dispensing of opioid analgesics (IRR: 1.06). n.s. reduction in the prescribing of opioid analgesics No change in buprenorphine utilization (IRR: 0.98).
		Alexandridis et al., (2019)	4 counties	Multifaceted community-coalition-based intervention featuring healthcare-provider training, expansion of addiction treatment	↓ community stigma toward people treated for OUD ↓ community stigma toward people treated for MOUD
		Davis et al., (2024)	1385 coalition stakeholders, representing 66 of 67 communities.	Communities That Heal (CTH), containing evidence-based practices, community engagement, and health communication campaigns	↑ access to therapies for OUD by 25 %
Social context, integration, reentry, and social supports	Social context, integration, reentry, and social supports	Glasgow et al., (2024)	67 communities in four states	CTH containing evidence-based practices, community engagement, and health communication campaigns	↑ development of community-level goals to address OUD
		Palombi et al., (2019)	Unknown	Grassroots coalition forums to increase awareness of overdose	↑ rate of abstinence ($X^2=20.0$) ↓ mental health and behavioral/social consequences($X^2=41.0$). n.s. increase in feeling socially connected from 90.0 % to 98.0 % ($X^2=2.67$).
		Crowthers et al., (2022)	117 PWOD	Peer recovery coaching featuring social supports	33.6 % received ongoing treatment services
		Dahlem et al., (2021)	122 patients who experienced non-fatal overdose	Community case management navigators and peer coaches	

SDOH Domain	Focus	Author(s) & Year	Sample Size	Intervention	Key Results
	<i>Family context</i>	Hyde et al., (2022)	43 previously incarcerated veterans	Post-incarceration peer-based intervention	↑ linkage and engagement in substance use (86 % vs. 19 %), mental health (93 % vs. 64 %)
		Shaffer et al., (2021)	79 individuals in Drug Court	Prosocial behavior intervention	↓ opioid use ↓ nights in jail
		Tuten et al., (2018)	135 individuals with OUD	Psychosocial treatment featuring social clubs	↑ abstinence at 6-months (slope: -0.56)
		Godwin and Conduct Problems Prevention Research Group, (2020)	713 at-risk kindergarteners	Social and emotional development, parenting, children's skills & behaviors	↓ opioid use later in life following a comprehensive childhood intervention
		Schaeffer et al., (2021)	98 PWOD	Comprehensive community treatment	↓ in days spent using opioids over time among parents (slope: -0.34) ↑ likely to report total past-month abstinence from all drugs (slope: -0.06) ↑ likely for past-month abstinence from opioids (slope: -1.45, p = 0.055)
Economic Stability	<i>Employment</i>	Crowthers et al., (2022)	117 PWOD	Peer recovery coaching	↑ rate of abstinence ($X^2=20.0$) ↑ employment ($X^2=19.17$)
		Holtyn et al., (2014)	98 PWOD	Employment-based reinforcement in therapeutic workplace for drug abstinence	92 % of participants engaged in methadone treatment 75 % negative opiate urine samples vs. 54 % in work reinforcement only group
		Novak et al., (2022)	91 PWOD	Abstinence-contingent wage supplements vs. TAU for drug abstinence and employment	↑ higher opiate and cocaine-negative urine samples in participants enrolled in contingent-wage vs. TAU (63.6 % of vs. 44.1 %)
	<i>Cash Gifts</i>	Yoo et al., (2022)	1000 low-income mothers with infants	Cash transfers to low-income mothers to influence substance use	n.s., differences in maternal substance use. n.s., opioid use effect size: -0.067 (
	<i>Basic needs</i>	Gryczynski et al., (2021)	400 patients with co-occurring SUDs	Patient navigators in hospitals featuring links to basic needs	Faster entry into treatment ↑ SUD treatment entry within 3 months (50.3 % intervention vs 35.3 % TAU) Faster entry into treatment ↑ MOUD treatment entry within 3 months (68 % intervention vs 40 % TAU) ↓ positive opioid drug test within 6 months (68 % vs 74 % TAU)
		Dahlem et al., (2021)	122 patients who experienced non-fatal overdose	Community case management navigators and peer coaches featuring linkages to basic needs	33.6 % received ongoing treatment services
		Tuten et al., (2018)	135 individuals with OUD	Psychosocial treatment featuring linkages to basic needs	↑ abstinence at 6-months (slope: -0.56)
	<i>Housing</i>	Crowthers et al., (2022)	117 PWOD	Peer recovery coaching	↑ rate of abstinence ($X^2=20.0$) ↑ housing stability ($X^2=7.14$)
		Dahlem et al., (2021)	122 patients who experienced non-fatal overdose	Community case management navigators and peer coaches featuring linkages to housing	33.6 % received ongoing treatment services
Neighborhood and physical environment					

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SDOH Domain	Focus	Author(s) & Year	Sample Size	Intervention	Key Results
		Davidson et al., (2014)	358 PW/OD	Consumer-participation for people with homelessness and substance use	↓ likely to report using stimulants or opiates at follow-up (OR: 0.17)
		Kelleher et al., (2021)	21 people at risk for OD	Housing first intervention, providing six months of rent, utilities, and OD prevention services	No participants reported opioid use by the 6-month follow-up
		Ly et al., (2024)	38 PW/OD	Pharmacist-led buprenorphine outreach services at Permanent Supportive Housing for people with OD.	↑ treatment adherence among intervention groups (37 % vs. 3 % achieved 80 % of prescription days filled)
		Tuten et al., (2018)	135 PW/OD	Access to recovery housing during treatment	↑ opioid abstinence at 1, 3, and 6 months (β -1.07, 1.46, 1.40) in the housed group. Employment ↑ abstinence at 3- and 6-months (β 1.20, 1.19)